



Bayes-MPC: Position Uncertainty and Navigation Trade-offs under Intermittent Pedestrian Observations

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Abstract

Occlusion makes crowd navigation a problem of controlling under missing information: a pedestrian that disappears from the sensor may still occupy the robot's intended path. Retaining a predicted position addresses this loss only partially, because the reliability of that prediction changes during an observation gap. This paper presents Bayes-MPC, a training-free navigation framework that combines recursive Bernoulli-Gaussian pedestrian beliefs with continuous trajectory optimization. Visible detections update track states, while missed detections propagate position uncertainty and modify existence confidence according to visibility evidence. The planner evaluates probability mass within robot-pedestrian collision regions and uses route-separated sampling to search for executable, goal-directed motion. The formulation consumes the posterior directly in trajectory evaluation rather than adding uncertainty features to a learned policy. In a frozen six-configuration confirmation with 600 new paired layouts per arm, the full method records 586 successes and 10 audited collisions, compared with 554 successes and 46 collisions when the same tracker supplies only posterior means. Retaining covariance while setting planning existence weights to one records seven collisions. A strengthened, development-selected duration margin also records 586 successes, with 11 collisions; its mean penalized time is 12.61 s versus 12.02 s for the full method. It does not exhibit more timeouts in this confirmation. Configuration-stratified results and separate visibility interventions distinguish the value of uncertainty from claims about density or human occlusion alone. EWMA and conformal alternatives remain competitive, including in retained historical severe-noise experiments. These results support probabilistic treatment of occluded tracks without establishing universal Bayesian superiority or a closed-loop safety guarantee.

Keywords

Crowd navigation; Partial observability; Bayesian filtering; Model predictive control; Collision probability; Occlusion

1 INTRODUCTION

Autonomous robots operating among pedestrians must reconcile collision avoidance, timely arrival, and predictable motion. These objectives become more difficult when the robot cannot continuously observe everyone nearby. One pedestrian can hide another, a previously visible person can disappear behind an obstacle, and an apparently open route can become occupied before the next reliable observation. A navigation algorithm must therefore reason not only about where a pedestrian is predicted to move, but also about how much confidence should be placed in that prediction.

Reactive geometric methods and learned crowd-navigation policies address different parts of this problem. Reciprocal collision avoidance constructs locally admissible velocities using a model of how other agents respond. Learned policies and value-based methods encode interaction patterns from experience, using pairwise representations,

attention, or recurrent state. These approaches provide important baselines, but a missing observation remains a substantive modeling decision: the controller can discard the pedestrian, extrapolate a remembered position, or maintain a distribution over the pedestrian's possible state. The last option is useful only if the planner converts that distribution into better closed-loop decisions.

This distinction motivates the present work. A more accurate predictive density is not by itself a navigation contribution. A controller may react identically to two different uncertainty estimates, a conservative risk region may prevent useful progress, or a restricted trajectory family may contain no suitable evasive motion. Consequently, prediction quality, uncertainty representation, and control generation must be evaluated together. The relevant question is not whether a Bayesian model can be attached to a navigation system, but whether its posterior provides information that a controller can use beyond a deterministic

memory or a calibrated safety margin.

We study this question through a training-free architecture. A recursive Bernoulli–Gaussian filter maintains one belief for each associated pedestrian. The posterior contains an existence probability, a mean position and velocity, and a covariance. During occlusion, the track continues to evolve rather than disappearing from the planner’s input. A continuous model predictive controller then evaluates the probability mass inside candidate collision regions and searches several motion families separately. It applies the first control of an evaluated trajectory, rather than averaging distinct evasive maneuvers into a new, unevaluated motion. The framework requires neither imitation learning nor reinforcement learning, and does not use a learned navigation policy.

The Bayesian component is a state estimator and the uncertainty source for planning, not a claim of mathematical exclusivity. Other probabilistic estimators and calibrated prediction sets can represent uncertainty. Moreover, in the clean observation model used by part of our benchmark, the covariance after an exact detection is determined by the subsequent time without observations. This makes a duration-dependent geometric margin an especially important alternative. A meaningful evaluation must give such simple alternatives enough development data to operate competitively.

The independent confirmation supports a narrower conclusion than universal dominance. Across six configurations, the full method reduces audited collisions from 46 to 10 relative to the same tracker supplying only means. Covariance with planning existence weights fixed to one records seven collisions. A strengthened age-margin control matches the full method’s 586 successes out of 600 and has three rather than four timeouts, although its penalized time is longer. Thus, the new evidence does not support making prevention of freezing the central advantage over all calibrated margins. Historical memory and severe-noise experiments are retained as separate evidence, not relabeled as new confirmation. These findings distinguish useful position uncertainty from a claim that every Bayesian component is indispensable or superior to a simpler alternative.

The main contributions are as follows.

1. A fully specified occlusion-belief-to-control formulation integrates recursive pedestrian state estimation, collision-region probability evaluation, and route-separated continuous trajectory optimization. Its assumptions and fallback behavior are stated explicitly.
2. Controlled closed-loop experiments separate current sensing, deterministic memory, posterior means, position covariance, and existence weighting, identifying which parts of the representation account for the observed navigation improvement.
3. Calibration-matched alternatives, a duration-margin control, and single-process timing measurements characterize the method’s practical benefits and limitations.

The formulation builds on established filtering, collision-probability, and sampling-based optimization tools. The study focuses on their integration and the closed-loop contribution of posterior information under occlusion. Section 2 reviews related approaches, and Sections 3–5 present the method and evaluation.

■ 2 RELATED WORK

■ 2.1 Reactive and Learned Crowd Navigation

ORCA [1] distributes collision-avoidance responsibility between interacting agents and constructs velocity constraints from their relative geometry. Its reciprocity assumption differs from the non-responsive-human setting considered here: when pedestrians do not account for the robot, the robot must compensate for that asymmetry. This does not make reciprocal methods invalid baselines, but the interaction assumption must be disclosed.

Deep value-learning methods [2] evaluate the consequences of candidate actions using learned interaction values. Recurrent representations support variable numbers of surrounding agents, while SARL aggregates human–human and human–robot interactions through attention [3]. DSRNN [4] structures recurrent computation around interaction relations. Prediction-aware interaction graphs further connect anticipated pedestrian motion to navigation. These methods motivate the retained-checkpoint benchmark in this paper, but that benchmark does not isolate the effect of Bayesian inference: both the state representation and the controller differ from Bayes-MPC.

Our principal attribution experiments therefore keep the MPC fixed and intervene on the information supplied to it. The comparison with older learned methods is useful as a navigation reference, whereas the controlled interventions address why posterior uncertainty matters within the proposed architecture.

■ 2.2 Occlusion, Belief, and Object Persistence

People-as-Sensors [5] learns an occlusion-aware representation from observable crowd behavior and incorporates it into navigation. Our simulator’s shadow-grid construction is ported from its sensing geometry; this is inherited infrastructure, not a contribution. PaS uses a learned latent representation and a reinforcement-learning policy to exploit visible-agent behavior as evidence about occluded space. Here the associated track must already have been detected: we propagate its state without learning a social inference model. The present filter only adjusts existence at its predicted location after a missed detection; it does not condition the spatial density on all observed-free cells. Thus spatial negative-evidence filtering is not an implemented contribution of this paper. This work shows that using belief in partially observed crowds is not itself a new research direction. Our implementation instead maintains explicit parametric tracks and consumes their posterior in a non-learned MPC. Its strongest existing evidence concerns pedestrians that were observed and subsequently occluded. It does not establish recovery of never-observed pedestrians from visible-agent behavior.

The Bayesian filtering machinery follows established recursive state-estimation principles [6, 7]. A missed measurement is not equivalent to evidence of absence: an occluded track should generally be propagated, while a track predicted inside a reliably observed empty region can receive negative detection evidence. Existence probabilities have also been considered in motion planning for automated driving. Consequently, neither object persistence nor multiplying collision risk by existence confidence can be claimed as unprecedented in isolation.

Table 1 Technical positioning against close families. The distinctions describe the formulations, not measured superiority. Shared-inference planner adaptations are distinct from end-to-end comparisons.

Method	Uncertainty or interaction representation	Control construction	Distinction relevant here
PaS	Learned occlusion representation	Learned navigation policy	Training and policy consumption differ
T-MPC++	Predicted dynamic obstacles with topology guidance	Parallel continuous optimization	Topological search is a strong planning comparator
SH-MPC	Sampled obstacle trajectories and a horizon-level risk formulation	Scenario-constrained MPC	Risk semantics and guarantees differ
SICNav	Explicit human interaction model	Bilevel MPC	Human response assumptions require matching
Bayes-MPC	Recursive associated-track means, covariance, and existence	Route-separated sampling with collision-mass evaluation	Controlled uncertainty attribution under occlusion

■ 2.3 Modern Probabilistic and Interactive MPC

Topology-driven parallel MPC [8] searches distinct trajectory classes to reduce the limitations of a single local optimization; T-MPC++ combines guided candidates with an additional unguided planner. This is a particularly relevant comparator because our route-separated search also attempts to preserve alternative passing motions. Our finite collection of seeded velocity populations is not equivalent to a certified enumeration of homotopy classes.

Scenario-based SH-MPC [9] addresses the probability of collision over a trajectory through scenario optimization. Its risk object differs from a per-time-step probability computed from independent marginal pedestrian beliefs. Giving the two planners numerically identical risk parameters would therefore not establish equal conservatism or equal guarantees. A meaningful operating-point comparison must examine measured safety and efficiency while retaining the distinction between their formal assumptions.

SICNav [10] embeds an interaction model in a bilevel planning problem. Its interaction assumptions and optimization structure differ from independent constant-velocity belief propagation. Our passive uncertainty propagation is neither active perception nor contingent belief-space planning. The current method neither optimizes expected information gain nor branches its future controls on hypothetical observations.

■ 2.4 Calibration and Sampling-Based Optimization

Bayesian filtering is sensitive to its observation and process models. A posterior can be internally consistent yet miscalibrated when pedestrians turn unexpectedly, identities are incorrect, or sensor noise differs from the assumed distribution. Conformal prediction [11] provides an alternative route to calibrated sets under its own assumptions. It is not intrinsically restricted to one global radius: nonconformity scores, conditioning variables, and adaptation mechanisms can produce different widths in different contexts. Our calibrated conformal and EWMA controls should thus be read as particular implementations, not exhaustive representatives of those families. Overlapping, closed-loop forecast residuals do not establish the exchangeability required by basic distribution-free coverage results; the radial calibration here carries no trajectory-level guarantee.

Cross-entropy trajectory optimization and sampling reuse have a substantial literature, including sample-efficient real-time planning. The proposed controller uses this established optimization family. The method-level question is how persistent uncertain tracks, probability-mass risk, and separate motion populations work together in partially observed navigation. The completed ablations support uncertainty-aware planning; a claim of a novel optimization principle would require additional evidence and is not made here.

■ 3 METHODOLOGY

■ 3.1 Problem Formulation and System Overview

The robot has planar position $p_t^r \in \mathbb{R}^2$, velocity $v_t^r \in \mathbb{R}^2$, radius r_r , and goal g . Pedestrian i has state $x_t^i = [(p_t^i)^\top, (v_t^i)^\top]^\top \in \mathbb{R}^4$ and radius r_i . Only a subset of pedestrians is detected at time t . Let Z_t be the associated detection set and $\mathcal{V}_t$ the robot's visibility information. Navigation seeks an executable trajectory that reaches g while controlling collision exposure and unnecessary delay.

Two visibility notions must be separated. Sensor occlusion determines which pedestrians the robot observes. The environment parameter `robot.visible=false` instead determines whether pedestrians account for the robot when generating their own motion. The latter is fixed to false in the completed benchmark. It does not imply that the robot itself receives no pedestrian observations.

The deployed information interface contains detections, observed velocities, radii, and visibility evidence, but not hidden ground-truth positions, pedestrian goals, or the true number of unobserved people. Ground truth is reserved for environment evolution and outcome auditing. The current simulator provides track identity association; handling identity switches is outside the validated scope.

■ 3.2 Recursive Pedestrian Beliefs

3.2.1 State and Existence Representation

Each maintained track is represented by

$$b_t^i = \{r_t^i, \mu_t^i, P_t^i, a_t^i\}, \quad (1)$$

where r_t^i is the probability of existence, μ_t^i and P_t^i are the conditional Gaussian state mean and covariance, and a_t^i counts missed observation steps. The Bernoulli track has empty-set probability $1 - r_t^i$ and

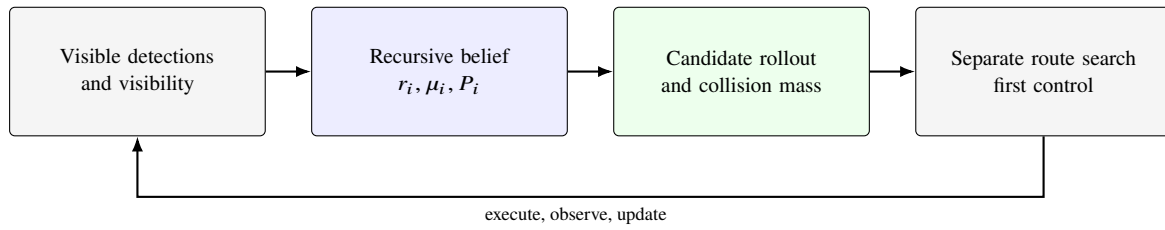


Fig. 1 Implemented belief-to-control pipeline. The feedback loop uses real subsequent observations. It is not a tree of hypothetical future observations, and no learned navigation policy is present.

singleton density $r_t^i \mathcal{N}(x; \mu_t^i, P_t^i)$. The collection is factorized across associated tracks. It is not a full posterior over correlated crowd trajectories or a multi-hypothesis data-association distribution.

The formal configuration creates tracks from visible detections. It does not enable a general spatial birth prior for never-observed people. Existence uncertainty therefore concerns the persistence and removal of previously observed tracks. This distinction is essential when interpreting the term *occlusion awareness*.

3.2.2 Prediction

For sampling interval Δt , the constant-velocity model is

$$F = \begin{bmatrix} I_2 & \Delta t I_2 \\ 0 & I_2 \end{bmatrix}, \quad Q = \sigma_a^2 \begin{bmatrix} \frac{\Delta t^4}{4} I_2 & \frac{\Delta t^3}{2} I_2 \\ \frac{\Delta t^3}{2} I_2 & \Delta t^2 I_2 \end{bmatrix}. \quad (2)$$

The predictive update is

$$\mu_t^{i,-} = F \mu_{t-1}^{i,+}, \quad P_t^{i,-} = F P_{t-1}^{i,+} F^\top + Q, \quad (3)$$

$$r_t^{i,-} = p_S r_{t-1}^{i,+}, \quad (4)$$

with survival parameter p_S . This model captures the growth of extrapolation uncertainty; it does not infer a hidden destination or assume that a pedestrian will negotiate with the robot. Unexpected turning or acceleration is represented only through the process-noise approximation.

3.2.3 Detection and Missed-Detection Updates

For an associated detection z_t^i containing position and velocity, the measurement matrix is I_4 and the noise covariance is

$$R = \text{diag}(\sigma_p^2, \sigma_p^2, \sigma_v^2, \sigma_v^2). \quad (5)$$

The Gaussian update uses

$$K_t = P_t^- (P_t^- + R)^{-1}, \quad (6)$$

$$\mu_t^+ = \mu_t^- + K_t (z_t - \mu_t^-), \quad (7)$$

$$P_t^+ = (I - K_t) P_t^- (I - K_t)^\top + K_t R K_t^\top. \quad (8)$$

The Joseph form maintains the covariance structure in finite precision. In the exact-observation condition, the implementation explicitly sets the state to the measurement and the covariance to zero. A confirmed detection resets the missed-step counter and sets existence confidence close to one.

When no associated detection is available, the predicted mean is queried against the visibility map. If this location should be observable but no detection occurs, negative evidence gives

$$r_t^+ = \frac{(1 - p_D) r_t^-}{1 - p_D r_t^-}. \quad (9)$$

If the predicted location is occluded or unobserved, this negative-evidence update is not applied. The update is a mean-location approximation: it does not integrate detection probability over every position in the Gaussian posterior. Tracks are removed after low existence confidence or a maximum missed age. These management thresholds are part of the model, not measurements of the true number of remaining pedestrians.

3.2.4 Prediction Across the Planning Horizon

For future step k , the planner receives

$$\mu_{t,k}^i = F^k \mu_t^i, \quad (10)$$

$$P_{t,k}^i = F^k P_t^i (F^k)^\top + \sum_{j=0}^{k-1} F^j Q (F^j)^\top. \quad (11)$$

The implementation uses the current existence weight over the candidate horizon; it does not repeatedly multiply this weight by p_S for every hypothetical step. The position mean and covariance are the corresponding upper blocks of (11).

After an exact observation, $P = 0$. With m missed steps, the future covariance satisfies

$$P_{m+k} = \sum_{j=0}^{m+k-1} F^j Q (F^j)^\top. \quad (12)$$

Under these assumptions, covariance is a function of observation age and prediction horizon, not a learned, person-specific measure of behavioral surprise. Equation (12) directly motivates the age-dependent margin baseline. Replacing this equation by an identical lookup table would be an implementation optimization, not a genuinely different non-Bayesian method.

3.3 Collision-Region Probability

3.3.1 Probability for One Pedestrian

Consider robot candidate position $p_{t,k}^r$ and pedestrian position distribution $\mathcal{N}(\mu_{i,k}^p, \Sigma_{i,k})$. Define the interaction radius

$$R_i = r_r + r_i + m_h, \quad (13)$$

where m_h is a geometric buffer shared across the matched planning interventions. The desired conditional collision probability is the Gaussian mass inside a disc:

$$q_{i,k} = \int_{\|y - p_{t,k}^r\| \leq R_i} \mathcal{N}(y; \mu_{i,k}^p, \Sigma_{i,k}) dy. \quad (14)$$

For isotropic position covariance $\Sigma_{i,k} = s_{i,k}^2 I_2$, the Gaussian collision-region calculation is a special case of established probabilistic collision evaluation [12]. Let $d_{i,k} = \|\mu_{i,k}^p - p_{i,k}^r\|$. The integral becomes

$$q_{i,k} = F_{\chi_2^2(\lambda_{i,k})} \left(\frac{R_i^2}{s_{i,k}^2} \right), \quad \lambda_{i,k} = \frac{d_{i,k}^2}{s_{i,k}^2}, \quad (15)$$

where $F_{\chi_2^2(\lambda)}$ denotes a noncentral chi-square CDF. As variance approaches zero, the implementation uses the corresponding deterministic overlap indicator.

The present process and measurement models produce isotropic position uncertainty under the evaluated initialization. Consequently, the scalar variance loses no spatial directional information in these experiments: the filter does not produce directional or multimodal position beliefs. If an arbitrary anisotropic covariance is supplied, the implementation replaces it by $s^2 = \text{tr}(\Sigma)/2$; (15) is then an isotropic approximation, not an exact anisotropic integral. General anisotropic collision integration is not claimed by this version.

Existence-weighted probability is

$$\pi_{i,k} = r_i^i q_{i,k}. \quad (16)$$

The effect of this weight is tested separately from covariance. It is possible for track management to use existence even when the planning weight is fixed to one; the two interventions should not be conflated.

3.3.2 Combining Pedestrians and Time

Under the factorized pedestrian approximation,

$$p_k = 1 - \prod_i (1 - \pi_{i,k}), \quad \ell_k = - \sum_i \log(1 - \pi_{i,k}). \quad (17)$$

The negative-log-survival quantity ℓ_k permits stable accumulation and yields $p_k = 1 - \exp(-\ell_k)$. Numerical clipping prevents logarithmic singularities. The factorization ignores correlated pedestrian motion and is therefore a modeling assumption rather than an unconditional safety certificate.

For one pedestrian with fixed positive isotropic variance and fixed existence, the collision probability decreases with distance from the predicted mean. A single-step probability threshold can therefore be represented by a radial distance condition (or an inactive constraint when the threshold exceeds the maximum probability). Probability constraints are not intrinsically beyond the expressive power of geometry. The tested differences instead involve the calibrated radius functions, aggregation across pedestrians, risk-dependent candidate costs, and their interaction with nominal swept checks. We consequently compare these complete evaluation rules rather than claim that all geometric alternatives must fail.

The risk limit varies over the horizon:

$$\epsilon_k = \epsilon_{\text{near}} + (\epsilon_{\text{far}} - \epsilon_{\text{near}}) \left(\frac{k-1}{H-1} \right)^\gamma, \quad k = 1, \dots, H. \quad (18)$$

The evaluated baseline uses $\gamma = 2$. Each step is checked against its own limit. Neither summing ℓ_k in the objective nor enforcing (18) computes the probability of any collision over the whole trajectory: successive pedestrian predictions are temporally correlated. In particular, these limits are not interchangeable with the horizon-level risk guarantee of SH-MPC.

■ 3.4 Continuous Trajectory Construction

3.4.1 Executable Motion and Swept Geometry

In the completed holonomic benchmark, controls are planar velocities $U = (u_0, \dots, u_{H-1})$ satisfying

$$\|u_k\| \leq v_{\text{max}}, \quad \|u_k - u_{k-1}\| \leq a_{\text{max}} \Delta t. \quad (19)$$

Candidate samples are sequentially projected onto these limits, then propagated by

$$p_{k+1}^r = p_k^r + \Delta t u_k. \quad (20)$$

The projection is included before trajectory scoring. The selected command is therefore scored in the same holonomic representation that is executed. The differential-drive comparison is a separate protocol because this equivalence does not automatically survive a conversion to angular and linear velocities.

Endpoint risk alone can miss a crossing between two simulation steps. The planner also computes a swept geometric clearance against the predicted pedestrian means. For relative endpoints d_0, d_1 , define $\delta = d_1 - d_0$ and

$$\alpha^\star = \text{clip} \left(-\frac{d_0^\top \delta}{\|\delta\|^2}, 0, 1 \right), \quad c = \|d_0 + \alpha^\star \delta\| - R_i. \quad (21)$$

Degenerate zero relative displacement uses $\alpha^\star = 0$. This check is geometric: it does not turn the endpoint probability model into a continuous-time stochastic collision guarantee. An independent swept-overlap audit is retained during evaluation.

3.4.2 Goal-Directed Objective

For a candidate trajectory, define distance to the goal region as $d_g(p) = \max(\|p - g\| - r_r, 0)$. The objective combines goal progress, control effort, smoothness, collision-related penalties, and failure to make progress:

$$\begin{aligned} J(U) = & w_g \sum_{k=1}^{H_a} d_g(p_k^r) + w_T d_g(p_{H_a}^r) + w_u \sum_{k=0}^{H_a-1} \|u_k\|^2 \\ & + w_\Delta \sum_{k=0}^{H_a-1} \|u_k - u_{k-1}\|^2 + w_{\text{prob}} \sum_{k=1}^{H_a} \ell_k \\ & + J_{\text{geom}}(U) + J_{\text{risk}}(U) + J_{\text{progress}}(U). \end{aligned} \quad (22)$$

Here H_a denotes the active portion of the horizon up to goal entry, if any. The goal-entry step remains subject to collision checks; steps after termination do not incur fictitious post-arrival interaction cost. J_{geom} penalizes overlap and insufficient clearance, J_{risk} penalizes excess over the scheduled probability limits, and J_{progress} discourages negligible progress. Their role is to shape candidate ranking; explicit feasibility labels are also used rather than relying solely on a finite weighted penalty.

3.4.3 Route-Separated Sampling

A single Gaussian over all useful controls can merge incompatible passing motions. For example, averaging left- and right-passing trajectories may produce a straight path through a pedestrian. To avoid

Algorithm 1 Recursive belief and route-separated receding-horizon control**Require:** Prior tracks, previous solution, detections Z_t , visibility $\mathcal{V}_t$

- 1: Predict each track using F, Q, p_S
- 2: Update detected tracks; apply justified missed-detection evidence
- 3: Remove expired tracks and initialize newly detected tracks
- 4: Predict surviving track means and covariances for H steps
- 5: Initialize separate warm-start, left, right, and waiting populations
- 6: **for** $j = 1, \dots, K$ **do**
- 7: Sample velocity sequences and project onto the robot limits
- 8: Compute rollouts, swept clearance, collision mass, and objective
- 9: Assign feasibility classes and update each population's elites
- 10: Retain the best evaluated sequence across all populations
- 11: **end for**
- 12: Execute its first control and shift the sequence for the next cycle

executing that average, we maintain separate sampling populations initialized from the shifted previous solution, left-biased motion, right-biased motion, and waiting-oriented motion. These are motion seeds, not a complete topological classification.

At each search iteration, temporally correlated perturbations are added to the population means. After projection and evaluation, elites are selected within each population. The sampling mean is smoothed toward its elites and each standard deviation is bounded below to retain exploration. The implementation uses short-range correlated perturbations; no claim of a learned diffusion model or long-horizon autoregressive noise process is made.

The best evaluated control sequence is retained across iterations and populations. The robot executes its first command. Distribution means are search parameters, not actions to be averaged across distinct successful routes. This design follows the broader sampling-MPC and cross-entropy family [13]; its incremental value relative to modern multi-route optimization remains a question for the independent comparator study.

3.4.4 Feasibility Hierarchy and Fallback

Candidates are prioritized in three groups. First, a trajectory that satisfies the geometric and probabilistic checks over the active horizon is preferred. Second, when none does, trajectories satisfying the first-step checks are ranked by horizon slack and objective value. Third, if no first-step-feasible candidate has been found, the controller prioritizes immediate nominal geometric clearance before applying the remaining ranking criteria.

The third group can violate the nominal risk constraints. It is a practical best-available-motion fallback, not a safety-preserving invariant set. Consequently, this algorithm should be described as risk-aware MPC, not as a recursively feasible or provably collision-free controller. Fallback frequency, feasibility class, and subsequent outcomes belong in the final failure analysis.

Table 2 Six-scene benchmark. Each completed six-scene cohort contains 100 episodes per scene, with equal weight in the macro average.

ID	Scene	Pedestrians	Spatial parameter
0	Baseline circle	5	Circle radius 4 m
1	Baseline square	10	Square width 10 m
2	Dense circle	10	Circle radius 4 m
3	Dense square	20	Square width 10 m
4	Large circle	12	Circle radius 6 m
5	Large square	20	Square width 14 m

3.5 Computational Structure and Scope

With B candidates, K iterations, H prediction steps, and M maintained pedestrians, the dominant pairwise evaluation scales as $O(KBHM)$. The implementation vectorizes this calculation, but the completed measurements use a desktop CPU path. The presence of a GPU in the host machine does not imply that the planner uses GPU kernels. There is no policy checkpoint to deploy; the executable artifact consists of the filtering and planning implementation, its numerical dependencies, and a frozen parameter configuration.

This formulation makes uncertainty operational: changing covariance changes collision-region probability and can change the selected trajectory. It does not make Bayesian inference logically indispensable. A calibrated set estimator or another predictive distribution can provide relevant information as well. The experiments test whether the particular recursive posterior is useful and which simpler alternatives remain competitive, rather than attempting to prove that only a Bayesian algorithm could solve the navigation task.

4 EXPERIMENTS

4.1 Research Questions

The evaluation distinguishes three questions. First, does retaining occluded pedestrians improve navigation relative to current sensing alone? Second, does position uncertainty provide useful information beyond a remembered or filtered mean? Third, can a simpler, fairly calibrated uncertainty representation achieve the same safety–efficiency tradeoff?

These questions require different comparisons. Older learned baselines test the complete navigation system; they cannot determine whether the gain comes from filtering or trajectory search. Posterior-mean and existence interventions address specific components. Calibration-matched alternatives test the necessity of the particular uncertainty treatment.

4.2 CrowdNav Benchmark

4.2.1 Scenarios and Physical Settings

The completed holonomic evaluation uses six configurations drawn from circle and square crossing geometries, summarized in Table 2. The six configurations vary crowd size and spatial extent; they are not six unrelated environment types. No corridor or merging experiment is implied by this table.

The control interval is 0.25 s, the time limit is 25 s, and the holonomic robot is limited to 1.0 m/s with acceleration bounded by 2.0 m/s². The benchmark robot and pedestrian radii are 0.30 m, and

the goal tolerance is the same 0.30 m: the simulator declares arrival when the robot centre enters a disc of one robot radius about the goal. The success radius entry of 0.25 m present in the environment configuration file is not read by the active environment and does not affect any reported outcome. The local visibility grid has 0.25 m resolution and spans approximately 5 m in each direction from the robot.

Pedestrian response to the robot is disabled (`robot.visible=false`). This is a deliberately demanding protocol, and it has a consequence that matters when comparing controllers: because pedestrians do not yield, holding still is not a safe action. A controller whose fallback on an infeasible or low-confidence step is to brake becomes a stationary obstacle that walking pedestrians can reach. Completion and collision rates under this protocol therefore penalise conservative fallback behaviour more sharply than an environment with reactive pedestrians would, and this should be kept in view when the numbers are compared with published results obtained under reactive crowds. The current-sensor condition supplies only visible detections, whereas remembered tracks can persist through subsequent observation gaps in the other conditions.

Figure 2 shows what this protocol removes from the robot's input. Two mechanisms are involved and they are not interchangeable. A pedestrian can be hidden behind another body, which is inter-agent occlusion proper, or can simply be farther away than the sensor reaches. Because these scenes contain no walls or static obstacles, every occluder is another pedestrian, so genuine occlusion requires the crowd to be dense enough for people to line up with one another. In the five-person scene of Figure 2 no pedestrian is occluded at all and the three missing ones are out of range, whereas the twenty-person square scene has five occluded and ten out of range. Statements in this paper about occlusion refer to limited-range sensing with inter-agent occlusion, and the belief maintains previously observed tracks through the resulting gaps; it does not place a prior on people who have never been seen.

The tested planner uses 16 prediction steps, 512 sampled sequences, four search iterations, and an elite fraction of 0.08. A geometric human buffer of 0.10 m is included. These settings are extracted from the completed experimental configuration, not from potentially different defaults in a later working tree. Early experiments using 1.2 m/s are not merged into the tables in this draft.

4.2.2 Calibration Rather Than Policy Training

The method does not undergo imitation or reinforcement learning. It nevertheless contains parameters and calibrated alternatives; training-free does not mean data-free or tuning-free. For the fair calibration study, each noise condition has 100 five-person development episodes, using case identifiers 3600–3699. Working-point selection uses development cases 3650–3699. Reuse within this development pool is allowed for model construction and selection, but its calibration coverage is not presented as an independent final test.

Clean observations have zero injected measurement noise. Severe observations add position standard deviation 0.10 m, velocity standard deviation 0.20 m/s, and detection probability 0.80. Methods receive the same noise-condition information and development trajectories. Testing a method configured for severe noise against an alternative cal-

ibrated only on clean observations would answer a different transfer question and is not the fair-noise comparison reported here.

4.3 Controlled Belief and Planning Interventions

Table 3 defines the ablation arms. With the exception of the explicit CV-memory control, the relevant uncertainty alternatives reuse the Bayesian track set and predicted means. This controls the nominal predicted trajectories while varying risk representation. It does not make the feasible sets identical: geometric-region controls change clearance buffers, whereas probability controls change probability penalties and constraints.

The CV-memory arm stores the last observation and extrapolates velocity with its own one-second expiry and visible-empty-space management. It is not identical to posterior mean. The posterior-mean intervention retains the Bayesian track selection but removes both covariance and planning existence weights. The existence-one intervention leaves existence-based track management intact. Consequently, it isolates the planner's use of existence, not the entire role of existence in the pipeline.

The age-margin control uses

$$m(a, k) = \kappa((a + k)\Delta t)^\eta. \quad (23)$$

Its coefficients are fitted from development residuals. The historical study used five target-coverage levels, 0.75, 0.85, 0.90, 0.95, and 0.975. The independent confirmation extends this development grid with 0.50 and 0.60 as specified below. The control removes covariance and existence inputs from the geometric planner to avoid unintentionally retaining the full probability mechanism. The fitted severe-noise margins are larger than the clean margins; this control does respond to the known noise condition through calibration, although not to online person-specific residuals.

Each family's working point is selected on development data using the registered criterion: among points whose empirical collision rate is within two percentage points of that family's minimum, choose the shortest failure-penalized time, with the smaller point index breaking ties. This supplies an explicit selection rule, but does not force all families to have the same true collision probability. The resulting selected-point comparison is therefore not automatically a matched-safety Pareto comparison. Small development samples also make point selection uncertain; the final independent table must retain its frozen selection.

4.4 Metrics and Statistical Analysis

4.4.1 Outcome and Time Definitions

Each episode is assigned one of three mutually exclusive audited outcomes. A recorded collision takes precedence over a nominal goal event. A success requires reaching the goal without an audited overlap. Remaining episodes are failures to finish within the deadline. The principal rates are success (SR), collision (CR), and timeout (TR). The stored collision-union and success-without-overlap flags define the completed tables. The independent final audit additionally reconciles those flags against the raw swept-overlap fields and environment event.

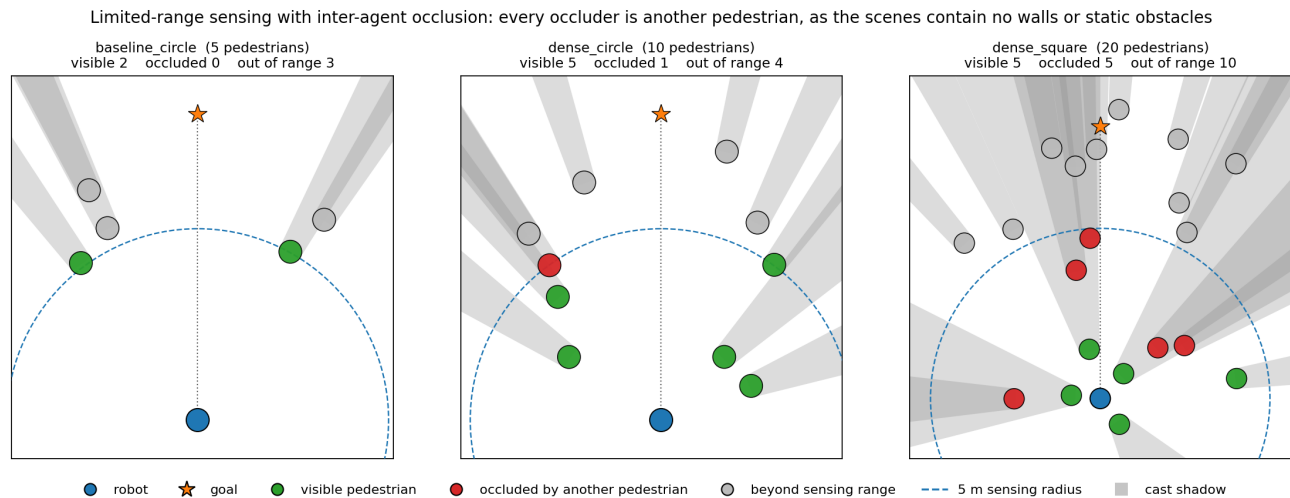


Fig. 2 What limits observability in this benchmark, drawn from three executed episodes. The shadows are the tangent-line polygons the simulator itself uses to decide visibility, and the visible/occluded split is the simulator's own. Two distinct causes are separated: a pedestrian hidden behind another body (red) and one simply farther away than the 5 m sensor reaches (grey). The scenes contain no walls or static obstacles, so every occluder is another pedestrian. In these illustrated frames, no pedestrian in the five-person scene is occluded and the three missing ones are out of range, whereas the twenty-person square scene has five occluded and ten out of range.

Table 3 Intervention definitions. A shared tracker does not imply identical uncertainty treatment. Conversely, the shared-tracker alternatives do not constitute a test of Bayesian state estimation against an entirely independent estimator.

Arm	Temporal state	Risk input	Question addressed
Current sensor	Current visible detections	Deterministic geometry	Value of retaining missing observations
CV memory	Independent CV extrapolation and expiry	Deterministic geometry	Value beyond simple memory
Posterior mean	Bayesian track means	No covariance or existence weight	Value beyond filtered positions
Existence-one	Same Bayesian tracks and means	Covariance; planning existence set to 1	Increment of planning existence weights
Full Bayes	Same Bayesian tracks and means	Covariance and existence	Complete implemented representation
Static covariance	Same Bayesian tracks and means	Calibrated visibility/horizon variance	Need for recursive covariance values
EWMA	Same Bayesian tracks and means	Residual-adapted covariance scale	Need beyond simple online adaptation
Conformal	Same Bayesian tracks and means	Calibrated geometric prediction regions	Probability mass versus calibrated sets
Age margin	Same Bayesian tracks and means	Duration/horizon geometric margin	Value beyond observation age

To avoid measuring speed only on the episodes a method happens to complete, define

$$T_p^{(e)} = \begin{cases} T_{\text{goal}}^{(e)}, & \text{audited success,} \\ T_{\text{max}}, & \text{otherwise,} \end{cases} \quad \bar{T}_p = \frac{1}{n} \sum_{e=1}^n T_p^{(e)}. \quad (24)$$

We call this *failure-penalized task time*, not average arrival time. It combines completion and speed: a lower value need not imply that successful trajectories are individually faster. Path length, mean progress, common-success paired arrival time, minimum clearance, and actual overlap steps are complementary measures for the final evaluation. An efficiency claim at equal safety requires safety non-inferiority or a pre-defined risk-budget comparison, not merely overlapping collision confidence intervals.

4.4.2 Pairing, Uncertainty, and Repeated Layouts

Methods are paired by scene and case identifier, with the same initial layout. Macro rates give each of the six scenes equal weight. With 100 episodes per scene, these equal the pooled rates. Binary comparisons use the exact paired McNemar test, based on discordant episodes, rather than an unpaired Fisher test. Time intervals use 4,000 paired bootstrap resamples within each scene. The layout is retained as the

comparison unit; individual control steps are not independent navigation trials.

The historical 5000–5099 layouts have been inspected and reused in later development. Their results remain useful as controlled descriptive evidence, but must not be relabeled as a newly held-out confirmation. The severe-noise cohort uses cases 5100–5699 and was independent of its own calibration at execution. It too becomes historical evidence once inspected. Reported historical p -values are exploratory and unadjusted. The final independent comparison must apply its registered primary contrast and multiple-comparison procedure, with no post-test changes to the selected working points.

Repeating planner seeds measures sensitivity to stochastic trajectory search. It does not create independent training seeds or new layout samples. Repeated runs of the same layout must remain clustered in any pooled uncertainty estimate. A non-significant difference does not demonstrate equivalence, and a small collision count can leave a practically important difference unresolved.

4.5 External Baselines

The completed historical full-observation benchmark includes ORCA, CADRL, LSTM-RL, SARL, DSRNN, and AttnGraph. Retained check-

points are evaluated in the local harness. They are not retrained under an equal optimization budget, and their original publication rates are not substituted for local outcomes. An apparent large gap therefore includes controller design, training provenance, and possible adaptation effects. The AttnGraph timeout pattern and the DSRNN occlusion-mask contract require particular care.

These retained policies are supplemental operating-condition references, not the main evidence for Bayesian inference and not a claim of superiority over current navigation methods. Separate T-MPC++ and SH-MPC adaptation experiments use a unicycle execution protocol and must not be pooled with this holonomic table. Sharing posterior inputs compares planner formulations, not independent inference systems. Their distinct trajectory-risk definitions also prevent equal numerical risk parameters from establishing equal conservatism. Solver failures and stationary collisions, where measured, describe the tested adapters and conditions; they do not prove a general failure boundary of the published methods.

Table 4 reports the repaired adapters rather than the superseded pre-repair numbers. These historical tests use 100 layouts per configuration and method under the continuous accelerating unicycle protocol. They are not additional holonomic confirmation samples. The selected T-MPC++ and SH-MPC horizons are 16 and 8, respectively, with their separately selected goal-task profiles and risk settings. Accordingly, the table is an operating-scope description, not an equal-formulation safety comparison.

Across configurations T-MPC++ reports native success on 16,873 of 21,598 steps; 308 of its 321 collisions end at low speed. In contrast, SH-MPC reports success on 35,763 of 35,769 steps. All 190 of its collision-terminal replies report success and nonzero slack. Its 122 low-speed collision intervals therefore cannot all be attributed to solver-failure braking. These are descriptive associations, not causal effects of stopping or guarantees that an unadapted published controller would fail. The common executor saturates 450 T-MPC++ requests; this is also retained in the supplementary record rather than described as zero requested bound violations. Synthetic acceptance included SH-MPC stopping before a static obstacle on the fixed straight reference and reaching the goal with a static detour reference. Reference construction, nonreactive pedestrians, slack handling, and task adaptation remain limitations. The compact per-episode extraction and source hashes are released as `external_scope.json`; the full-observation rows are retained there too.

■ 4.6 Frozen Independent Confirmation Protocol

The new confirmation separates development from testing. The 100 environmental seed identifiers 3001017–3001116 are each instantiated in all six configurations. Before execution, 30,107 historical result files containing 5,004 recorded case identifiers were inspected, and generated initial-state hashes were checked for overlap under each of the six generators. Independence here refers to this recorded history and generator; it is not a claim that all possible historical artifacts have been recovered.

The main comparison contains 600 paired layouts and six arms: pos-

terior mean, covariance with planning existence set to one, full Bayes, age margin, EWMA, and conformal regions. All use the original holonomic actuator and common search budget. Each arm controls its own robot and maintains its own resulting legal observation history, posterior, and warm start. Pairing fixes initial layouts and seed rules, not subsequent observations after the actions diverge. The Bayes working point has near/far limits 0.225/0.75 and probability weight 1. Other calibrated families retain their recorded development selection. To challenge the historical age-margin boundary selection, coverage targets 0.50 and 0.60 were added using only the original calibration observations and 50 original development layouts per target. Target 0.60 was selected: 50/50 successes and 10.19 s mean penalized time, compared with 10.53 s at the historical 0.75 target. The 0.50 target had one collision and 10.31 s penalized time. These additional 100 episodes are development data, not confirmation data. Working-point selection uses the original five-pedestrian circular development configuration. Parameters then transfer unchanged across the six test configurations; the study does not claim separately optimized dense-scene operating points for every comparator. This extension gives the geometric alternative additional development scrutiny; we do not claim identical tuning effort merely because the controller is shared.

For visibility attribution, the first 20 registered seeds per configuration are evaluated with posterior mean and covariance-one under range-only sensing and full current-state observation, adding 480 episodes to the 3,600 main episodes. The combined range-and-occlusion condition reuses the corresponding main results. Range-only sensing removes shadow polygons while retaining the same occupancy grid, field-of-view radius, and entity visibility convention. Full observation exposes current states, never future trajectories. Pedestrian behavior is unchanged. Although clean full observations reset the current covariance to zero, future prediction still accumulates process noise. Mean-only and covariance-aware control need not therefore have identical actions under full observation.

The primary endpoint is the paired audited collision difference between full Bayes and posterior mean. Confidence intervals resample 100 seed blocks, each containing all six configurations, with 10,000 bootstrap replicates. Secondary collision comparisons use blockwise sign randomization with Holm correction. Episode-level McNemar values, where provided, are descriptive because configurations sharing a seed are clustered. Visibility effects compare differences of method differences, not significance labels from separate tests. The fixed sample count is a resource commitment, not a guarantee of power for every small effect. Concurrent execution times are recorded but are not used as deployment latency evidence. The unchanged simulator emits its terminal timeout on the discrete control grid: the 25 new timeout episodes have a recorded terminal timestamp of 25.25 s and are charged the fixed 25 s failure penalty. No successful or colliding episode in this confirmation terminates after 25 s. This legacy endpoint convention is retained for all arms rather than silently changing the execution protocol.

Table 4 Historical repaired-adaptor scope under range limits and occlusion. Each row has 100 layouts. S/C/T are counts. Solve is the percentage of native solver replies reporting success; it is not a safety certificate. Low-speed C counts collision intervals ending below 0.1 m/s, divided by all collisions; it does not identify speed at first contact. Bayes uses sampling, so native solver success is not applicable. None of these rows contributes to the holonomic confirmation table.

Configuration (people)	Adapter	S/C/T	Penalty (s)	Solve (%)	Low-speed C
Baseline circle (5)	Bayes	97/3/0	10.785	–	1/3
	T-MPC++	80/20/0	13.650	93.33	20/20
	SH-MPC	81/16/3	15.745	100.00	11/16
Baseline square (10)	Bayes	83/17/0	12.668	–	4/17
	T-MPC++	50/46/4	18.010	76.68	41/46
	SH-MPC	50/19/31	20.548	99.98	12/19
Dense circle (10)	Bayes	83/17/0	14.395	–	3/17
	T-MPC++	57/43/0	18.455	82.44	41/43
	SH-MPC	54/20/26	20.675	99.99	13/20
Dense square (20)	Bayes	72/27/1	16.838	–	3/27
	T-MPC++	5/89/6	24.468	40.93	88/89
	SH-MPC	5/54/41	24.813	100.00	36/54
Large circle (12)	Bayes	95/4/1	16.110	–	0/4
	T-MPC++	51/49/0	21.085	82.89	48/49
	SH-MPC	54/22/24	22.505	99.96	11/22
Large square (20)	Bayes	83/17/0	13.408	–	2/17
	T-MPC++	25/74/1	21.528	70.86	70/74
	SH-MPC	20/59/21	23.270	99.98	39/59

5 RESULTS

5.1 Independent Configuration-Stratified Confirmation

Table 5 is the primary navigation table. Each row contains 100 new layouts, not repeated planner seeds. The two 20-pedestrian square configurations have different arena sizes, and the circular configurations also differ in size and count; these are configuration effects, not a controlled one-dimensional density experiment.

The full method records 586/600 successes (97.67%), ten collisions, and four timeouts. Mean-only planning records 554 successes, 46 collisions, and no timeouts. Cov-1 records 589 successes, seven collisions, and four timeouts. The difference is most visible in dense square crossings, but full Bayes is not the best uncertainty-aware arm in every configuration: age margin and Cov-1 both have fewer collisions and more arrivals than full Bayes in the dense square configuration.

The strengthened age margin records 586 successes, 11 collisions, and three timeouts overall. Its 12.605 s mean penalized time exceeds Bayes' 12.024 s, but the historical statement that this margin necessarily freezes more often does not replicate after the development extension. EWMA records 582 successes and 15 collisions at 11.948 s penalized time; conformal regions record 578 successes and 11 collisions at 12.829 s. These complete trade-offs are more informative than ranking all methods by success alone.

Table 6 confirms the primary collision reduction, with a block-randomization Monte Carlo $p = 5.0 \times 10^{-5}$ at the resolution of 20,000 draws. Full Bayes rescues 40 mean-only failures but changes eight mean-only successes into failures: five collisions and three timeouts. The complete three-by-three event transitions are retained in the released results. The direct Cov-1 minus mean collision effect is -6.50 percentage points with nominal 95% interval $[-8.50, -4.50]$, supporting position uncertainty without establishing an additional planning-existence benefit. Neither the secondary collision tests nor their intervals establish Bayes superiority over the calibrated uncertainty-aware

alternatives. The planning-existence intervention deserves an explicit negative finding: full Bayes has three fewer successes and three more collisions than Cov-1 overall, while their timeout counts are equal. This is not uniform across configurations: in baseline square full Bayes has one more success at the same collision count, whereas in dense square it has four fewer successes and three more collisions. The data neither justify a beneficial planning-existence claim nor establish that existence weighting is universally harmful. They do not remove the separate need for track management in either arm.

5.2 Separating Range Limits and Human Occlusion

Table 7 reports the fixed 120-layout visibility subset. Adding covariance reduces collisions by seven in the combined condition, five with range limits alone, and two under full current-state observation. The combined-minus-range difference of collision effects is -1.67 percentage points with 95% seed-block interval $[-4.17, 0.00]$; combined-minus-full is -4.17 points with interval $[-8.33, 0.00]$. Both intervals include zero. The point estimates are consistent with larger uncertainty value under restricted visibility, but this subset does not establish that human occlusion amplifies the effect beyond range restriction or that the benefit is exclusive to occlusion.

5.3 Mechanisms and Failure Boundaries

We replay recorded actions for the first registered Bayes layout in each configuration. Figure 3 selects, by an outcome-independent rule, the reported track with the most missed-observation steps. The plots show propagation and observation resets, including departures beyond sensing range. The model radius is not an empirical coverage guarantee, and a long-lived distant track need not be decision-relevant.

For a separate same-state diagnostic, steps 8 and 24 of those six layouts supply 12 legal belief snapshots. At each snapshot, mean, Bayes, and age evaluate the same cold-start pool of 512 bounded candidates. Eight snapshots produce a different first command for Bayes versus

Table 5 Frozen independent confirmation under combined 5 m sensing and human occlusion. Counts are successful arrival (S), audited collision (C), and timeout (T); each configuration–method row contains 100 episodes. Cov-1 retains covariance but fixes planning existence to one. Age uses the strengthened 0.60 development target. No external policy or unicycle-planner result is mixed into this table.

Configuration	Pedestrians	Method	S	C	T	Penalized time (s)
Baseline circle	5	Mean	99	1	0	9.61
		Cov-1	100	0	0	9.82
		Bayes	100	0	0	9.82
		Age	100	0	0	10.06
		EWMA	100	0	0	9.76
		Conformal	100	0	0	10.10
Baseline square	10	Mean	93	7	0	10.49
		Cov-1	96	3	1	10.60
		Bayes	97	3	0	10.47
		Age	97	2	1	11.16
		EWMA	95	5	0	10.46
		Conformal	95	3	2	11.44
Dense circle	10	Mean	95	5	0	12.01
		Cov-1	100	0	0	11.49
		Bayes	100	0	0	11.56
		Age	100	0	0	11.72
		EWMA	100	0	0	11.48
		Conformal	99	1	0	11.95
Dense square	20	Mean	76	24	0	14.63
		Cov-1	94	3	3	13.70
		Bayes	90	6	4	14.10
		Age	95	3	2	14.52
		EWMA	90	7	3	13.90
		Conformal	87	5	8	15.32
Large circle	12	Mean	99	1	0	14.71
		Cov-1	100	0	0	14.87
		Bayes	100	0	0	14.81
		Age	100	0	0	15.33
		EWMA	100	0	0	14.85
		Conformal	100	0	0	15.54
Large square	20	Mean	92	8	0	11.18
		Cov-1	99	1	0	11.44
		Bayes	99	1	0	11.40
		Age	94	6	0	12.86
		EWMA	97	3	0	11.23
		Conformal	97	2	1	12.63

mean; four do not. These are not the actual warm-start CEM searches, and a different command alone does not imply better navigation. Figure 4 shows both preselected dense-square states; all 12 states and candidate selections are released, rather than only the favorable examples.

Table 8 retains failures as well as improvements. A read-only seven-by-seven Gaussian quadrature diagnostic also finds mixed observed-free and unknown support in 13.3–23.8% of hidden-track steps across configurations. This diagnostic does not measure decision-relevant risk changes, separates neither moving shadows nor growing support, and provides no evidence for a new shape-changing negative-evidence algorithm. No such algorithm is introduced or selected on these test layouts.

■ 5.4 Historical Evidence for Persistent Memory

The following historical tables use the earlier, repeatedly inspected layouts. They document how the study was developed and supply com-

parisons not rerun in the confirmation, especially current sensing and independent CV memory. Their counts must not be pooled with Table 5 or described as new held-out evidence. Historical age-margin results use the former 0.75 target, not the strengthened 0.60 target in the new primary table.

Table 9 provides the central controlled comparison. Discarding a pedestrian when it becomes unobservable results in 67.17% success and 32.83% collision. Constant-velocity memory increases success to 93.00% and reduces collision to 7.00%, showing that much of the difference from the sensor-only controller is attributable to retaining information across observation gaps.

The full posterior improves further, reaching 97.17% success and 2.00% collision. Relative to CV memory, this is a 4.17 percentage-point increase in success and a 5.00 percentage-point reduction in collision. The collision count falls from 42 to 12, a 71.4% relative reduc-

Table 6 Paired full-Bayes minus comparator effects on the new 600 layouts. CR differences are percentage points; time differences are seconds. Intervals are nominal 95% seed-block bootstrap intervals. Secondary collision p -values are Holm-adjusted; the mean-only comparison is the preregistered primary test. An interval containing zero does not establish safety equivalence.

Comparator	CR difference [95% CI]	Collision p	Time difference [95% CI]	Rescued / harmed
Mean	-6.00 [-8.00, -4.00]	5.0×10^{-5}	-0.081 [-0.393, 0.224]	40 / 8
Cov-1	0.50 [0.00, 1.17]	0.983	0.039 [-0.056, 0.140]	1 / 4
Age	-0.17 [-1.33, 1.00]	1.000	-0.582 [-0.802, -0.355]	8 / 8
EWMA	-0.83 [-2.17, 0.33]	0.983	0.076 [-0.121, 0.260]	8 / 4
Conformal	-0.17 [-1.33, 0.83]	1.000	-0.805 [-1.023, -0.599]	15 / 7

Table 7 Visibility intervention on the same 120 layouts per row. S/C are success/collision counts; all rows have zero timeouts. Cov-1 uses the same frozen parameters in every observation condition.

Observation	Method	S / C	Time (s)
Range + occlusion	Mean	111 / 9	12.160
	Cov-1	118 / 2	12.110
Range only	Mean	113 / 7	11.898
	Cov-1	118 / 2	11.981
Full current state	Mean	116 / 4	11.510
	Cov-1	118 / 2	11.969

Table 8 Failure diagnostics over the new 600 layouts per arm. Hidden means any maintained track was unobserved at collision, not necessarily the colliding pedestrian. Degraded means the first-step-infeasible choice at that time. Stalled timeouts contain a two-second interval with less than 0.10 m net goal progress; these are descriptive co-occurrences, not causal attribution.

Method	Collisions	Hidden	Degraded	Stalled TR
Mean	46	44	29	0
Cov-1	7	6	3	4
Bayes	10	9	5	4
Age	11	7	10	3
EWMA	15	14	9	3
Conformal	11	9	10	11

tion. There are 37 paired episodes in which CV memory collides and the full method does not, against seven in the opposite direction. The exact paired collision p -value is 5.30×10^{-6} . The paired bootstrap interval for the collision-rate difference is $[-7.00, -2.83]$ percentage points.

This safety improvement should not be rewritten as an unconditional speed improvement. Failure-penalized task time increases from 12.091 s to 12.220 s: a paired difference of 0.129 s, with a 95% bootstrap interval of $[-0.182, 0.439]$ s. The point estimate is approximately a 1.1% time increase. The full method has five timeouts where the CV-memory arm has none. The evidence supports substantially fewer

Table 9 Controlled occlusion ablation, six scenes and 600 layouts per arm. The audited collision indicator takes precedence over the environment outcome. All arms use the same MPC configuration; CV memory has its own track-management rule. These historical layouts have subsequently been reused, not newly held out.

Method	SR (%)	CR (%)	TR (%)	$\bar{T}_p$ (s)
Current sensor	67.17	32.83	0.00	15.33
CV memory	93.00	7.00	0.00	12.09
Posterior mean	93.50	6.50	0.00	12.03
Covariance, existence=1	96.67	2.67	0.67	12.26
Bayes-MPC	97.17	2.00	0.83	12.22

Table 10 Clean six-scene duration-margin comparison after development-only point selection. Each arm has 600 episodes on reused layouts. Non-significant collision differences do not establish safety equivalence.

Method	SR (%)	CR (%)	TR (%)	$\bar{T}_p$ (s)
Bayes-MPC	97.17	2.00	0.83	12.22
Age margin	92.67	1.67	5.67	14.13

collisions with a small observed time difference, not simultaneous statistically established improvement in every metric.

5.5 Historical Covariance and Existence Interventions

The posterior-mean arm achieves 93.50% success and 6.50% collision. Because this arm retains the Bayesian track set and predicted means, its difference from the full method is not explained simply by placing more remembered positions in the planner. The full method reduces collisions from 39 to 12 ($p = 2.53 \times 10^{-5}$) and increases success by 3.67 percentage points ($p = 0.00126$). Its penalized-time difference is +0.191 s, with interval $[-0.123, 0.510]$ s.

Retaining covariance while fixing all planning existence weights to one yields 96.67% success and 2.67% collision. This recovers most of the observed difference between posterior mean and the full method. The remaining four-collision difference between existence-one and full Bayes has exact paired $p = 0.289$; the 0.50 percentage-point success difference has $p = 0.549$. Thus, the current data most clearly support position-uncertainty-aware planning. They do not establish a significant additional navigation benefit from existence weighting.

This finding has two implications for the contribution statement. First, the Bernoulli-Gaussian formulation is an accurate description of the implemented state, but the empirical claim should not put all its emphasis on the Bernoulli component. Second, setting a planning existence weight to one does not remove the filter's existence-based track management. A claim that existence modeling is altogether unnecessary would require a separate, properly matched management intervention.

5.6 Historical Age-Margin Operating Point

Equation (12) makes age-dependent uncertainty a strong simple explanation of the clean-condition result. Table 10 tests a calibrated geometric margin with this information. At the independently selected development points, age margin attains 92.67% success, 1.67% collision, and 5.67% timeout. The full method attains 97.17%, 2.00%, and 0.83%, respectively.

Age margin has two fewer collisions, a difference not resolved by the paired test ($p = 0.804$). However, it has 27 fewer successes ($p =$

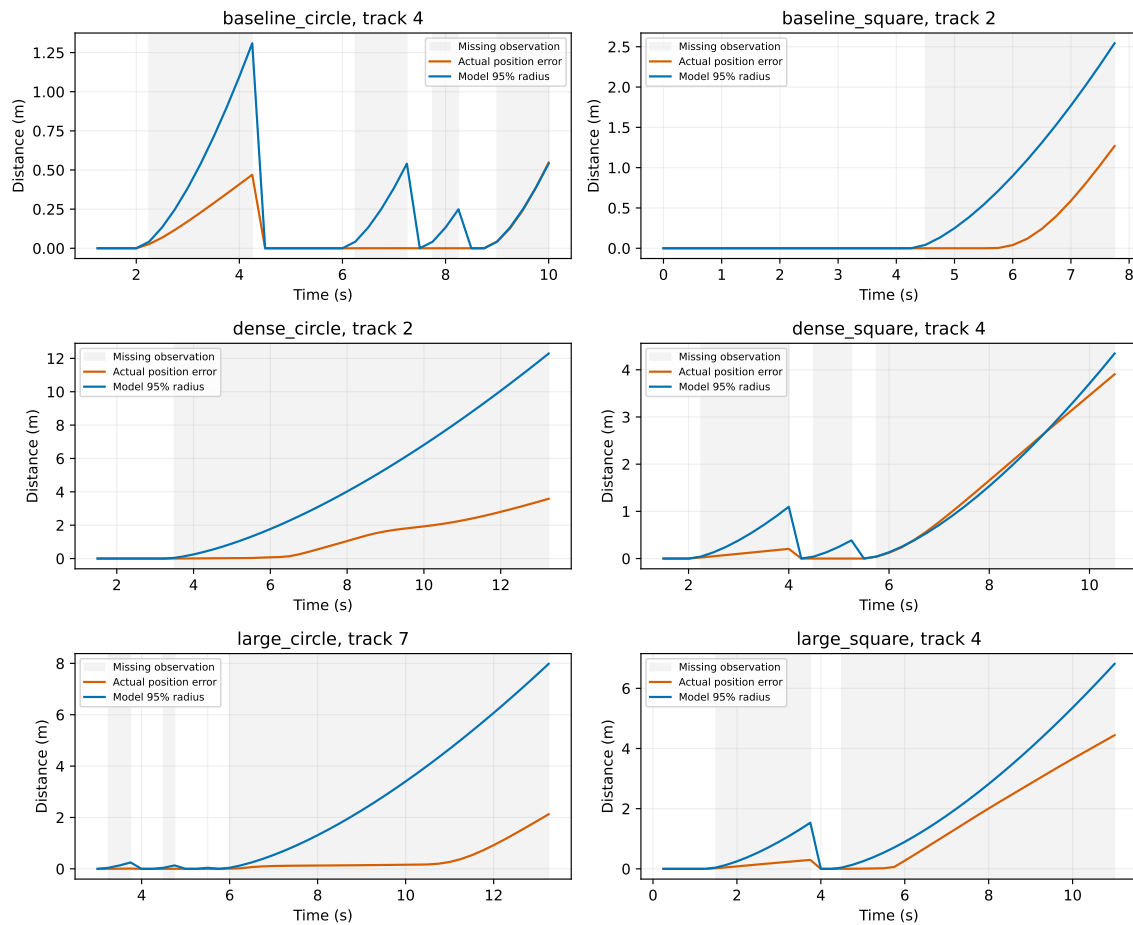


Fig. 3 Recorded-action replay in all six configurations. Shading marks missing observations; blue is the current Gaussian's nominal 95% position radius and orange is actual position error, used only for retrospective scoring. Selection uses missed-observation duration, not prediction quality. This is not a population calibration test.

9.85×10^{-5}) and 29 additional timeouts ($p = 1.31 \times 10^{-7}$). The full method reduces failure-penalized task time by 1.912 s, with a paired 95% interval of $[-2.207, -1.617]$ s for full minus age margin. In the dense-square scene, success is 92% for the full method and 73% for age margin.

The selected geometric control is therefore more prone to freezing in dense conditions. The result is consistent with an advantage from evaluating local collision mass instead of treating the whole enlarged region as an obstacle. It does not prove equal safety: absence of a significant collision difference is not a safety-equivalence test. Nor does one fitted power-law margin establish that every deterministic duration-aware construction must fail. The supported statement is that this calibrated, budget-matched control does not recover the full method's completion and penalized-time performance.

■ 5.7 Historical Severe Observation Noise

Table 11 changes both observation quality and the calibration protocol. All five methods receive severe-noise development data before their working points are frozen. The final evaluation comprises 600 dense-square layouts per arm, not a six-scene macro average.

The full method achieves 74.00% success, 11.67% collision, and

Table 11 Matched-information severe-noise test: 600 layouts per arm, dense square only. Calibration and point selection use five-person development episodes. Position noise is 0.10 m, velocity noise 0.20 m/s, and detection probability 0.80.

Method	SR (%)	CR (%)	TR (%)	$\bar{T}_p$ (s)
Bayes-MPC	74.00	11.67	14.33	18.13
EWMA	75.67	13.00	11.33	17.89
Conformal	74.17	11.00	14.83	18.73
Static covariance	63.83	35.33	0.83	17.46
Age margin	49.67	6.83	43.50	21.62

14.33% timeout. EWMA achieves 75.67%, 13.00%, and 11.33%; conformal achieves 74.17%, 11.00%, and 14.83%. The full method's collision differences against EWMA and conformal have paired p -values of 0.410 and 0.747. The corresponding success p -values are 0.423 and 1.000. These results do not support a clear full-posterior advantage over either calibrated alternative in success or collision.

Failure-penalized time is 18.129 s for the full method, 17.885 s for EWMA, and 18.726 s for conformal. Full minus EWMA is +0.244 s, with interval $[-0.100, 0.590]$ s. Full minus conformal is -0.598 s, with interval $[-1.010, -0.188]$ s. The latter exploratory time difference does not establish a matched-safety Pareto improvement because the

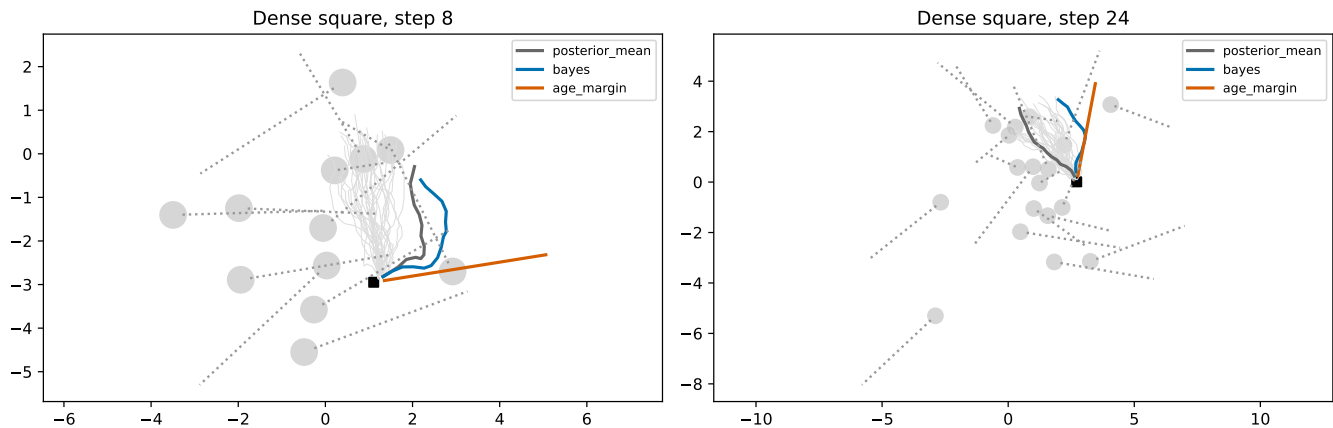


Fig. 4 Common-candidate diagnostics at both fixed dense-square times. Colored curves are selected robot trajectories; faint curves show a subset of the common candidate pool. Dotted lines and circles are legal pedestrian predictions and reported positions, not future ground truth. Selection changes demonstrate use of uncertainty but are not additional closed-loop outcomes.

Table 12 Historical fully observed same-controller comparison (600 layouts per arm). External checkpoint rankings are excluded from the principal Bayesian attribution.

Method	SR (%)	CR (%)	TR (%)	$\bar{T}_P$ (s)
Bayes-MPC	98.33	0.83	0.83	12.10
Static covariance	98.33	0.83	0.83	12.71

safety comparison remains uncertain.

Static covariance produces 35.33% collision and only 0.83% timeout. Its low timeout rate is not evidence of better risk handling: much of its task failure occurs through collision instead. Compared with this control, the full method reduces collision by 23.67 percentage points but increases timeout by 13.50 percentage points. This is a materially different operating pattern. It also cautions against reporting only one failure category.

Age margin produces the lowest observed collision rate, 6.83%, but times out in 43.50% of episodes and succeeds in only 49.67%. Its conservative behavior is not simply the absence of any noise adaptation: its severe-noise calibration uses larger margins than its clean calibration. The difficulty is that a duration-dependent inflated region cannot express the same local probability tradeoff as the other tested risk representations.

The most important inference from this experiment is negative but informative: fair calibration substantially narrows the apparent Bayesian advantage. EWMA and conformal remain viable alternatives. A previous comparison in which one method used noise-aware settings and another retained clean calibration would not isolate the uncertainty algorithm itself.

■ 5.8 Historical Full-Observation Baselines

Under full observation, the completed Bayes and static-covariance arms both achieve 98.33% success and 0.83% collision. Their equal aggregate rates do not imply equal trajectories or actions. The respective failure-penalized times are 12.096 s and 12.712 s, so even this summary shows different behavior. The claim of pointwise equivalence under full observation is not supported.

Table 13 Planner-repeat stability on reused six-scene layouts, with 600 episodes per row. Two complete planning repetitions are reported; these are not independent training seeds.

Method	Seed	SR (%)	CR (%)	$\bar{T}_P$ (s)
Bayes-MPC	0	97.17	2.00	12.220
Bayes-MPC	1	96.50	2.50	12.334
Posterior mean	0	93.50	6.50	12.029
Posterior mean	1	91.17	8.83	12.299

The larger historical checkpoint ranking is retained in the archived study record, not used as the main comparison in this manuscript. Training distributions, controller designs, and adapter assumptions differ. Published percentages obtained under another speed, geometry, termination rule, or visibility setting cannot resolve these differences. The present evidence therefore does not claim broad superiority over ORCA or learned navigation policies.

■ 5.9 Historical Planner-Seed Sensitivity

The additional complete locally available planning repeat yields 96.50% success and 2.50% collision for the full method, compared with 91.17% and 8.83% for posterior mean. This is consistent in direction with the original repeat, but two complete repeats are not evidence of arbitrary seed robustness.

■ 5.10 Historical Single-Process Timing

The single-process timing audit concatenates per-step timing arrays before computing percentiles. Averaging per-episode 95th percentiles would produce a different, generally incorrect estimate of the overall 95th percentile. This historical audit uses point index 3 for all calibrated families, rather than every family's subsequently selected setting. In particular, its age margin is not the new 0.60 operating point. It is evidence about the recorded execution costs, not a timing comparison of all newly selected confirmation arms. The full method has a planning p95 of 163.8 ms and pipeline p95 of 198.2 ms over 1,284 measured steps. Its pipeline p99 is 203.7 ms, and no measured step exceeds the 250 ms control interval.

Table 14 Completed single-process desktop timing audit. Percentiles concatenate raw step arrays, not episode percentiles. Each arm has 20 episodes on one dense-square setting; no hard real-time or onboard claim is implied.

Method	Steps	Plan p95	Pipe p95	Miss (%)
Current sensor	610	50.0	51.1	0.00
CV memory	784	65.0	66.8	0.00
Posterior mean	765	68.5	103.2	0.00
Age margin	2020	60.7	90.9	0.00
Conformal	1333	65.6	97.5	0.00
Static covariance	1007	139.5	169.5	0.00
EWMA	1080	139.3	169.6	0.00
Bayes-MPC	1284	163.8	198.2	0.00

The full method is computationally heavier than the alternatives in this audit: posterior mean has a pipeline p95 of 103.2 ms, conformal 97.5 ms, and CV memory 66.8 ms. Thus, computational efficiency should not be confused with shorter navigation time. The evidence supports operation within the measured desktop control period for these 20 dense-square episodes, not an unconditional real-time guarantee or onboard performance claim.

Concurrent batch runs showed more deadline misses, illustrating the importance of resource contention. Their timings are not merged with the single-process measurements in Table 14.

■ 5.11 Discussion and Limitations

The independent confirmation strengthens the evidence that treating extrapolated positions as certain can be harmful in this controller. It does not establish that every Bayesian component is needed: Cov-1 has fewer observed collisions than the full method, and uncertainty-aware alternatives remain competitive. Historical comparisons against a separate CV memory provide complementary, but no longer independently held-out, evidence about track persistence.

The new margin calibration changes the interpretation of freezing. The selected 0.60 target matches Bayes' overall success count and has one fewer timeout. Its longer penalized time is an efficiency trade-off, not proof of equal safety or a universal freezing disadvantage. Dense square results also show that the full method does not uniformly dominate calibrated geometric alternatives. The severe-noise evidence continues to limit any claim of Bayesian superiority over EWMA or conformal regions.

The evaluation is limited to the tested simulation dynamics, association assumptions, and calibrated uncertainty families. The matched-tracker alternatives isolate risk representation but do not compare entirely independent perception systems. End-to-end reproduction of contemporary planners under their original deployment conditions and transfer to a physical platform are outside the completed evaluation; Table 4 concerns local shared-posterior adaptations only. The current isotropic filter can be represented by an age-dependent covariance lookup in the clean condition. The experiments therefore support using its uncertainty in control, not the logical impossibility of a non-Bayesian implementation with the same inputs and equations. No direction-aware negative evidence model, active sensing policy, or new filtering theorem is claimed.

A post-confirmation development audit replays all 600 Bayes layouts, recovering 27,978 control states. Removing all hidden tracks whose current Gaussian spans both observed-free and unknown space would reduce the selected plan's collision probability by at least one percentage point at some prediction step in 17.84% of states. Gaussian cell-mass integration and set-inclusion checks bracket this potential-relevance fraction at 16.47–20.80%. This is not an implemented belief update, an observed constraint switch, or a navigation benefit. A separate single-frame conditioning diagnostic on the 12 previously fixed snapshots leaves all first commands unchanged in its finite candidate pools; only four snapshots activate the spatial update, and one fails the numerical refinement tolerance. These limited developmental observations neither establish a spatial-shape benefit nor exclude one on other states or under recursive filtering. They do not alter the frozen confirmation results or justify a new direction-aware method claim.

■ 5.12 Evidence and Reproduction

The confirmation protocol and executable experiment sources were frozen in commit 12ec42b; the result archive records file hashes rather than relying on filenames alone. `confirmation_summary.json` supplies Tables 5 and 6, `visibility_summary.json` supplies Table 7, and `failure_analysis.json` supplies Table 8. `external_scope.json` supplies Table 4 and includes per-episode repaired-adaptor diagnostics and source hashes. The released `episodes.jsonl` contains all 4,080 registered outcomes, actions, step timings, and audit records. Development episodes remain separate. The two mechanism figures have corresponding trace or candidate-snapshot records. `historical_manifest.json` maps the retained historical tables to their original episode files and source snapshots; these are not additional test data. The supplementary `binding/` directory contains the separately registered developmental relevance audit, cell-integration check, and spatial-conditioning probe, including their numerical limitations. These are not new test episodes.

Code and the compressed evidence bundle are available in the repository branch `occlusion-paper-confirmation-20260913` at <https://github.com/jinglongjiang/bayes-occ-mpc>. The bundle includes calibration inputs, original path mappings, manuscript source, and figure assets. Its read-only portable analysis verifies input and code hashes before regenerating statistics and recorded-action replays. The tested scope is relocation on the same installed environment, not a separate hardware or dependency installation. Different RVO2 builds require a separate numerical validation.

■ 6 CONCLUSION

This paper has presented a training-free crowd-navigation framework that maintains explicit beliefs over intermittently observed pedestrians and consumes those beliefs in continuous trajectory optimization. Its recursive state representation preserves occluded tracks, while collision-region probability and route-separated search translate uncertainty into executable motion. The formulation does not depend on a learned temporal backbone or a trained policy.

The new six-configuration confirmation records 46 audited collisions for posterior-mean planning, ten for full Bayes, and seven for

covariance with planning existence weights fixed to one. Full Bayes reaches the goal in 586 of 600 layouts. The strengthened age margin reaches the same number of goals and does not time out more often, although it takes longer in the penalized-time measure. Thus, uncertainty relative to a point prediction is the supported mechanism; universal superiority over calibrated margins or adaptive covariance is not.

The present evidence supports uncertainty-aware planning through occlusion, not universal Bayesian superiority, guaranteed safety, or completed sim-to-real transfer. Future work will examine contemporary continuous-planning baselines, differential-drive deployment, track-association errors, and the interaction between prediction dispersion and finite-horizon risk.

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